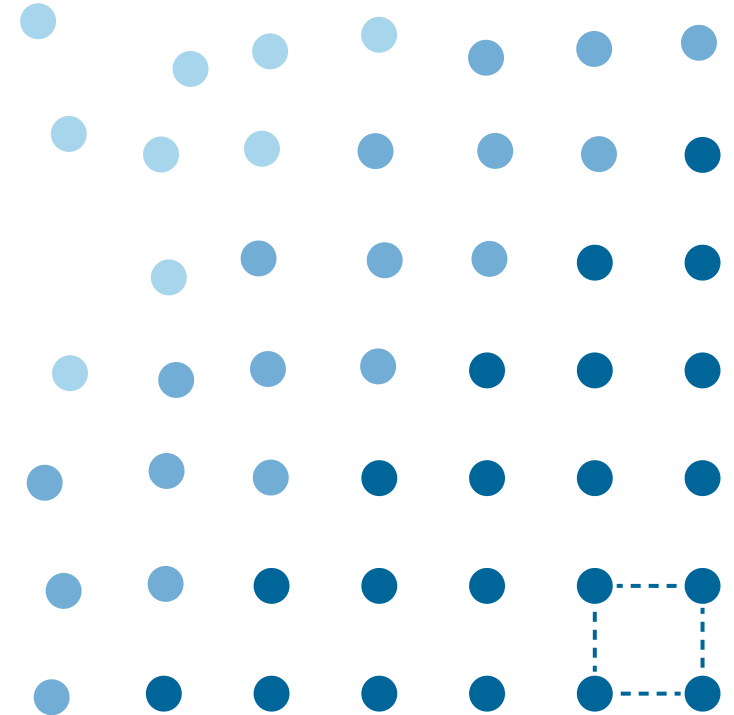


Recovering crystal symmetry in noisy structures

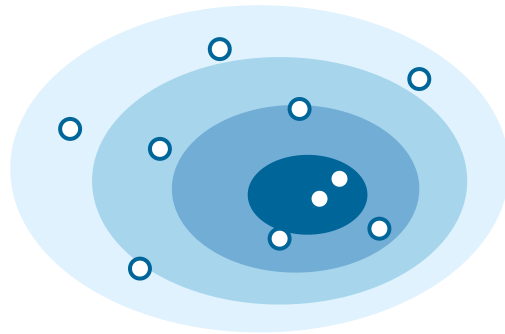
A machine learning approach

Alessandro Serafini

Group Meeting · 1 October 2026



Where the problem comes from

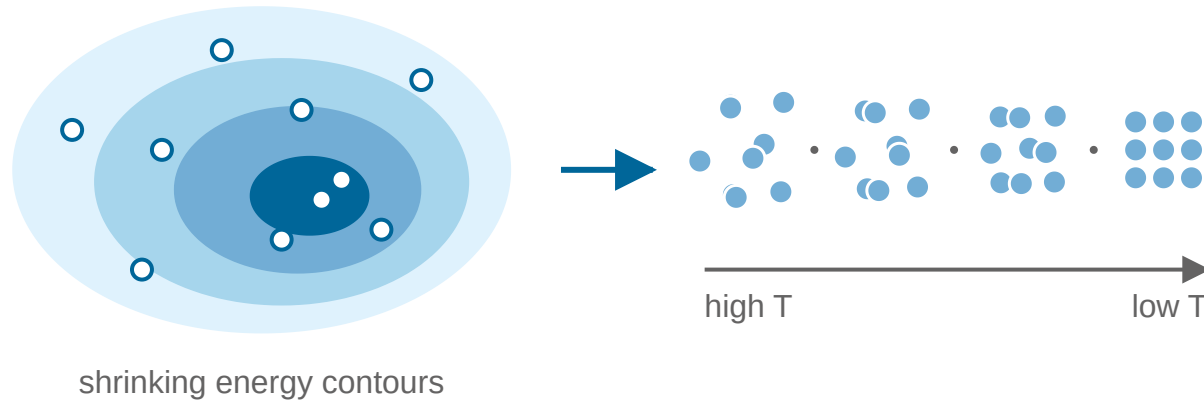


shrinking energy contours

Nested Sampling

Explores the configuration space; direct access to the partition function over a wide T range

Where the problem comes from



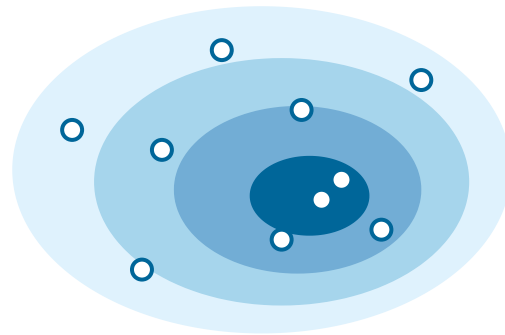
Nested Sampling

Explores the configuration space; direct access to the partition function over a wide T range

Trajectory of structures

Many structures, each representative of the system at a given temperature

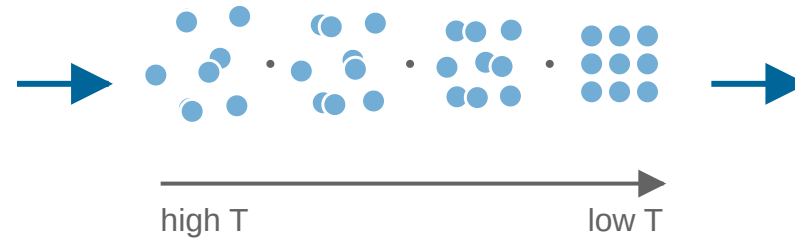
Where the problem comes from



shrinking energy contours

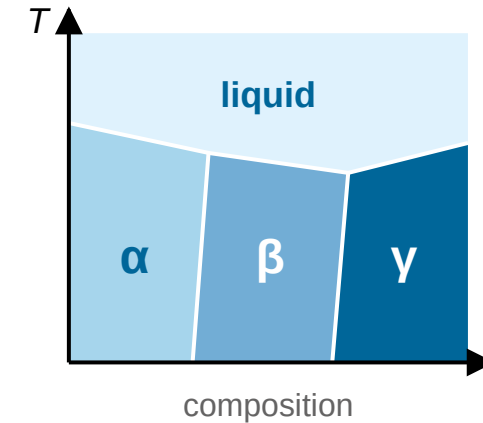
Nested Sampling

Explores the configuration space; direct access to the partition function over a wide T range



Trajectory of structures

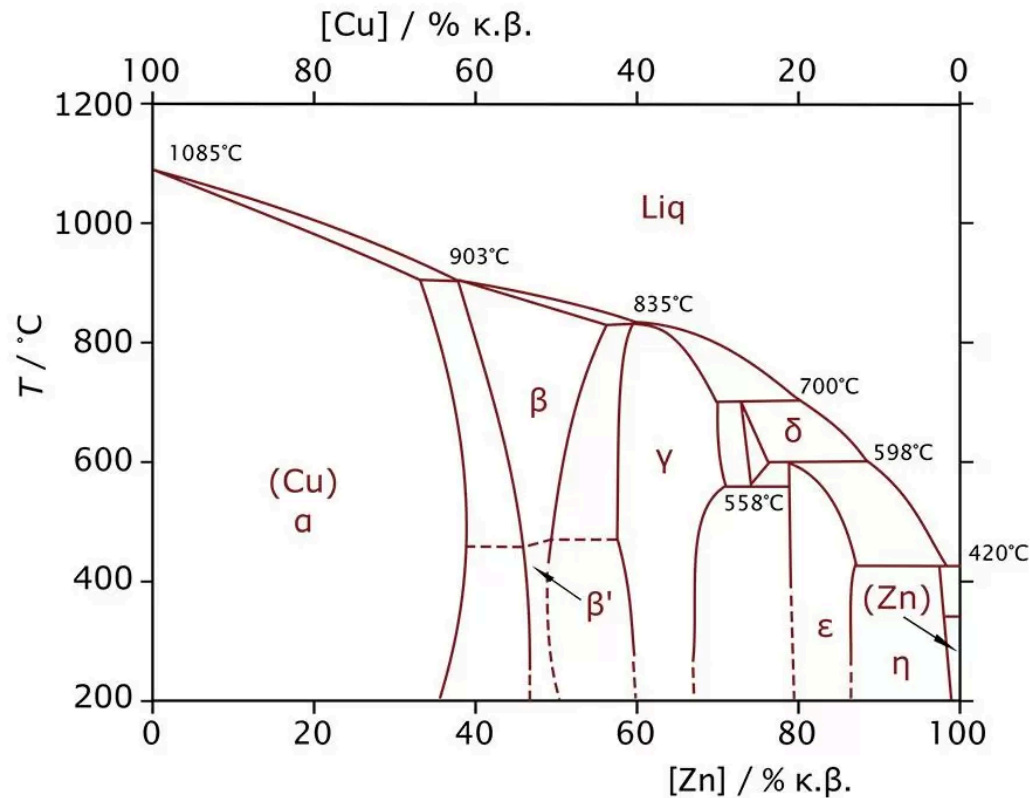
Many structures, each representative of the system at a given temperature



Phase diagram

Goal: one phase label, and its symmetry, for each region of T and composition

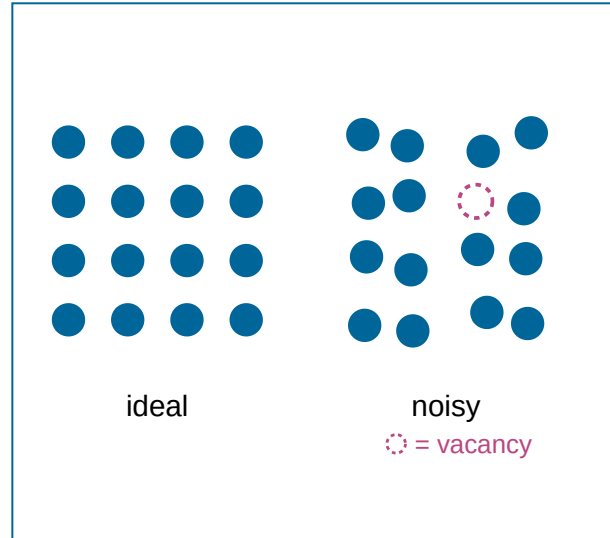
Example: the Cu–Zn (brass) phase diagram



Guiding example

Brass, with phases $\alpha / \beta / \gamma$, each with its own crystal symmetry — motivation only, not used throughout the talk

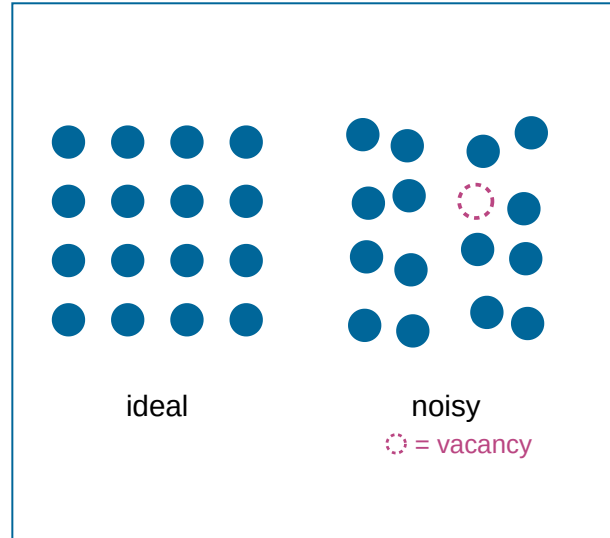
Why a classical symmetry finder is not enough



Noisy structures

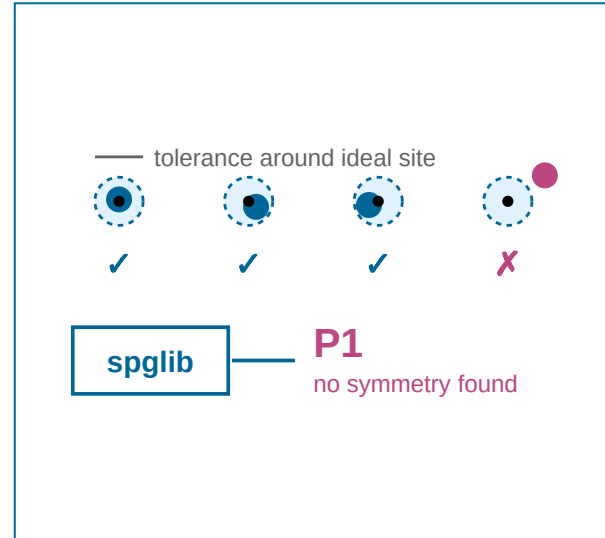
Monte Carlo at finite T : displaced atoms, vacancies, some far from any local minimum of the PES

Why a classical symmetry finder is not enough



Noisy structures

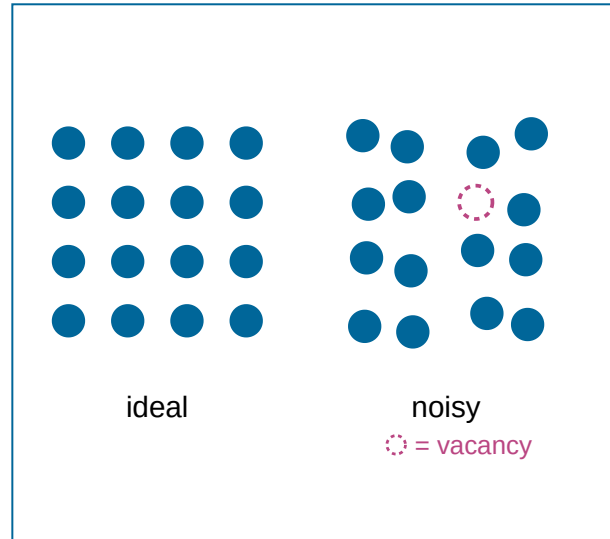
Monte Carlo at finite T: displaced atoms, vacancies, some far from any local minimum of the PES



Classical tools fail

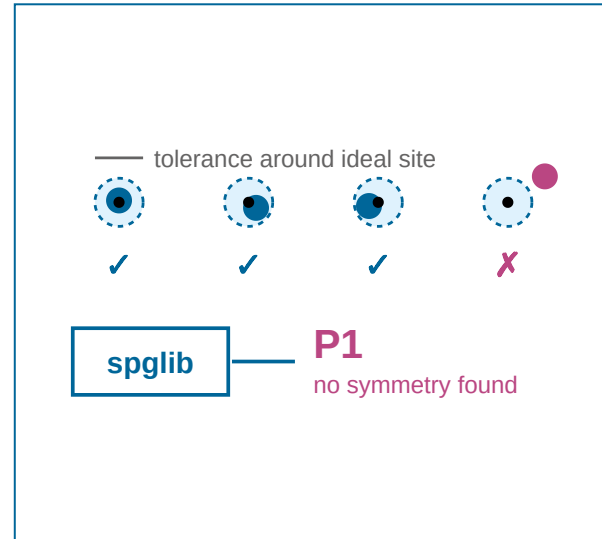
Symmetry finders like spglib need every atom within a small tolerance of its ideal site

Why a classical symmetry finder is not enough



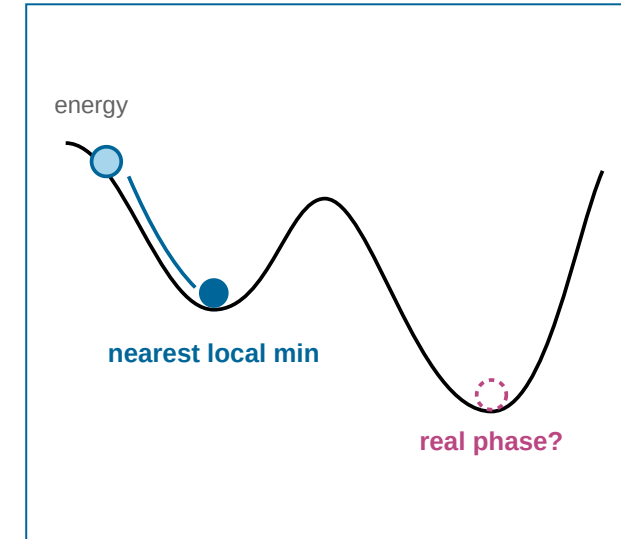
Noisy structures

Monte Carlo at finite T: displaced atoms, vacancies, some far from any local minimum of the PES



Classical tools fail

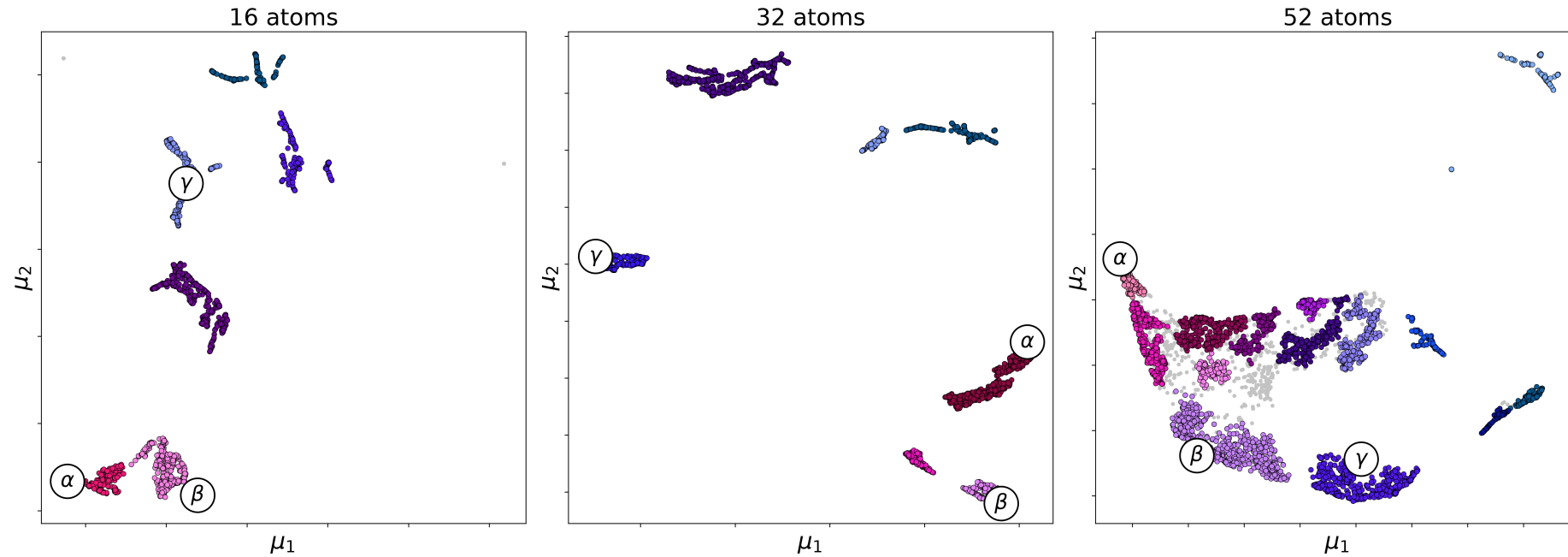
Symmetry finders like spglib need every atom within a small tolerance of its ideal site



Local optimisation is not enough

Costly (only a subset), and it stops at the nearest minimum — no guarantee it is the real phase

First attempt: SOAP + UMAP



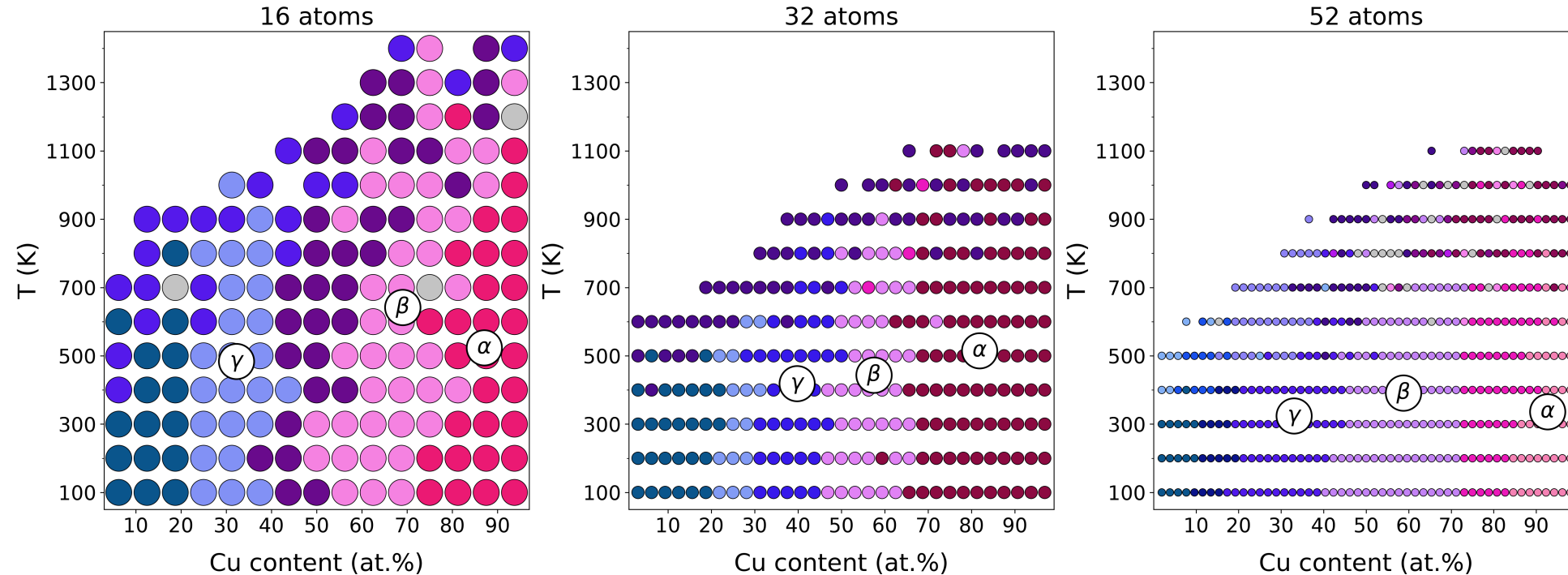
SOAP → UMAP

Snapshots from MD runs started from representative Nested Sampling structures → SOAP descriptors → 2D map with clear clusters

Reference phases

Brass α / β / γ from the literature, projected into the same space, fall inside those clusters

First attempt: clusters back in T–composition space



Back to T–composition

Colouring each (T, composition) point by its cluster gives coherent regions, consistent with the α / β / γ reference phases

What if we had no reference structures?

General

*knowledge of
crystal symmetries*

What if we had no reference structures?

General

*knowledge of
crystal symmetries*

Transferable

*independent of the
chemical system*

What if we had no reference structures?

General

*knowledge of
crystal symmetries*

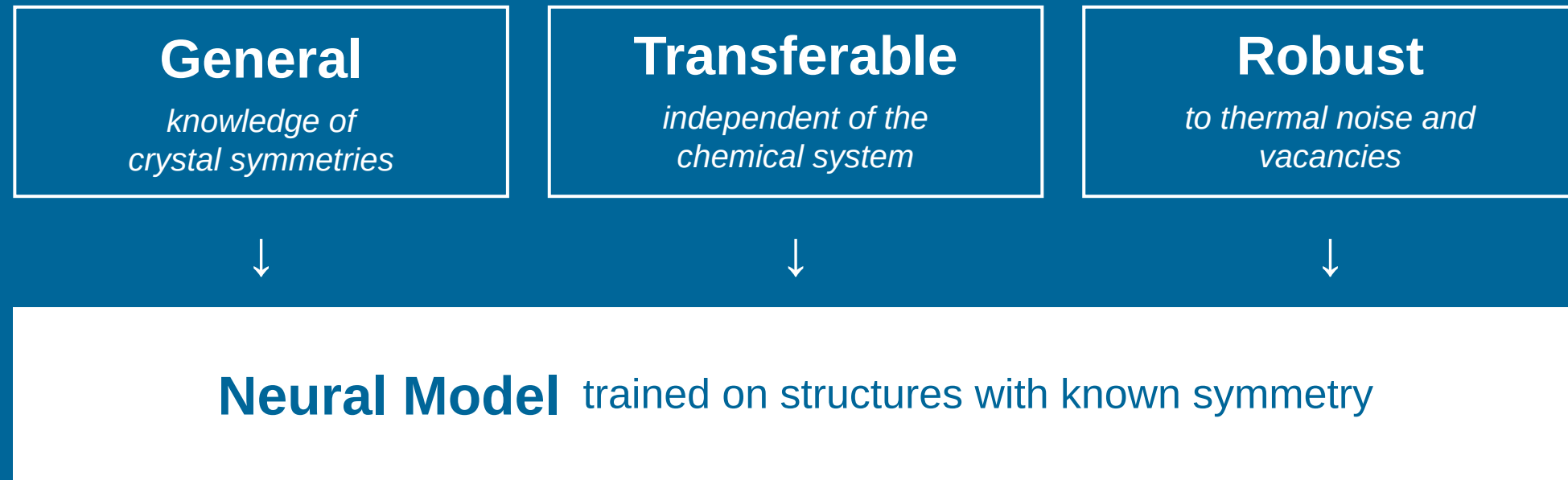
Transferable

*independent of the
chemical system*

Robust

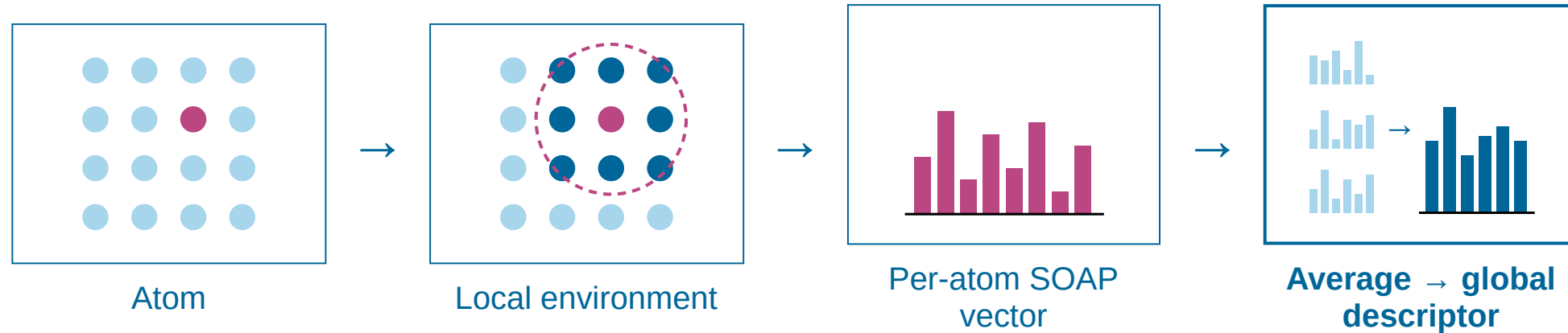
*to thermal noise and
vacancies*

What if we had no reference structures?



A global descriptor for each structure

Simplest case: single-element systems



Invariant to

Permutation

the order of the atoms does not matter

Rotation

of the whole structure

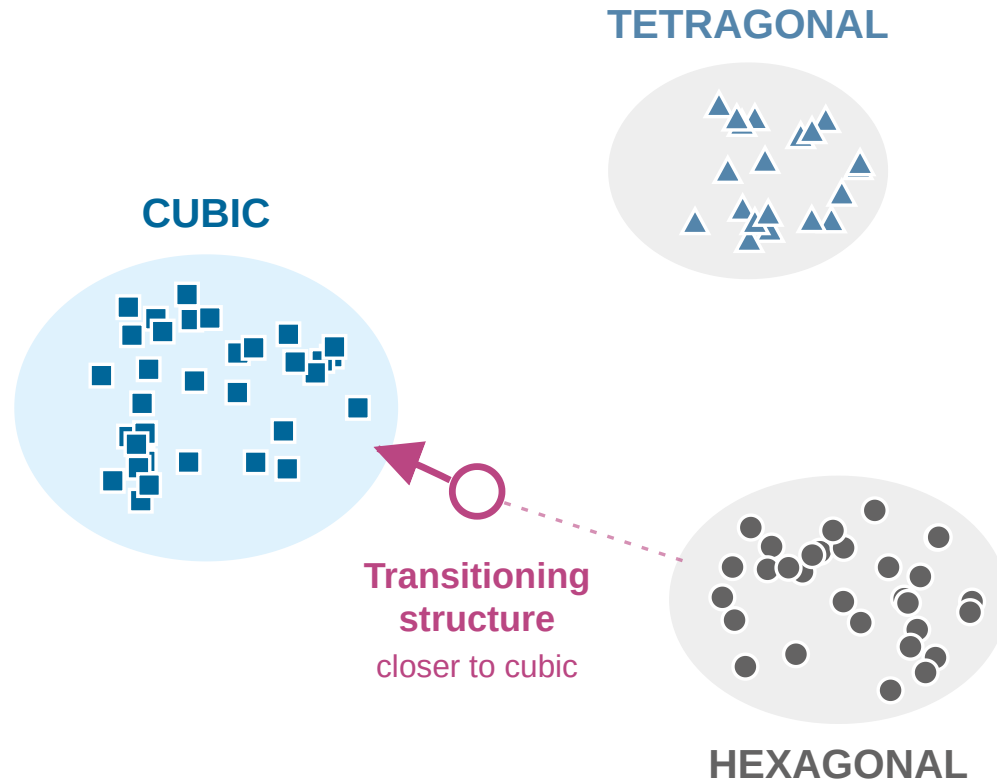
Translation

of the whole structure

Choice of cell

the periodic cell used to represent it

An explorable low-dimensional space



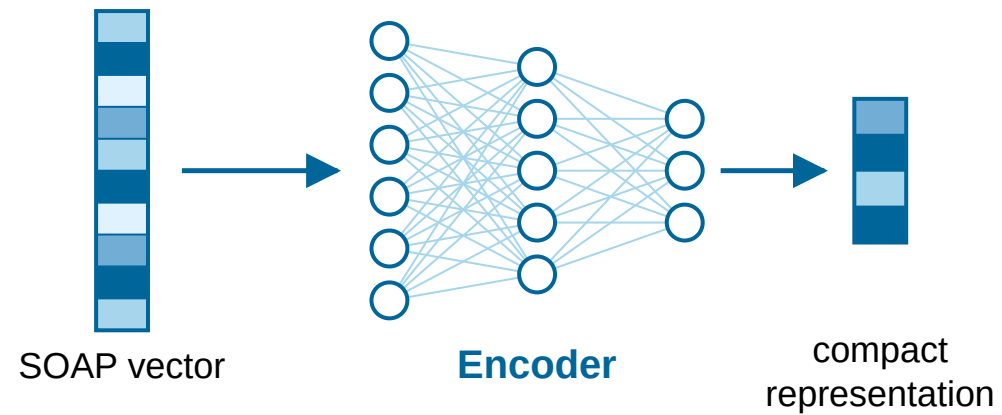
Not just a hard label

We also want to see where a structure sits in the space

Useful at the edges

In transition regions or ill-defined phases: which symmetry does it tend towards?

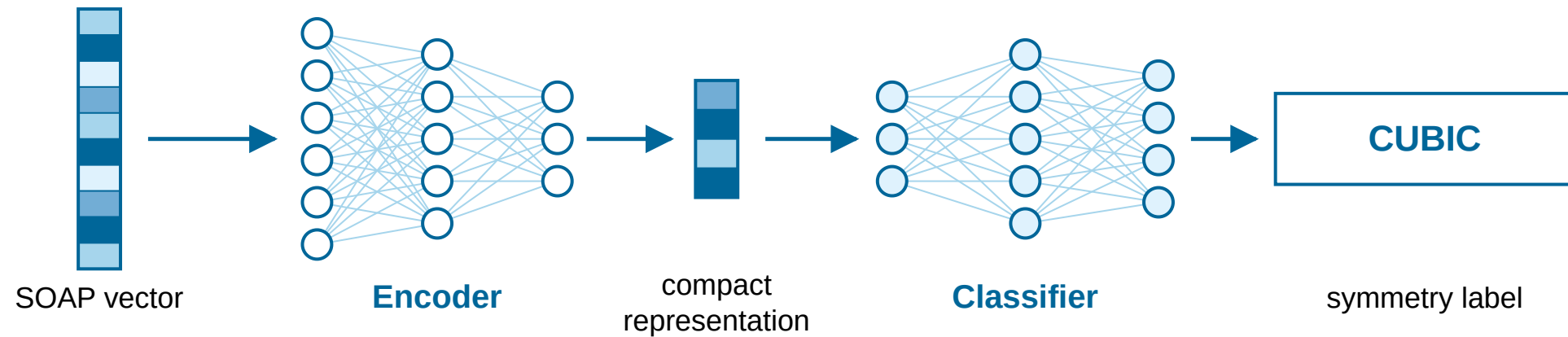
What we need: encoder + classifier



Encoder

Reduces the structure to a compact representation

What we need: encoder + classifier



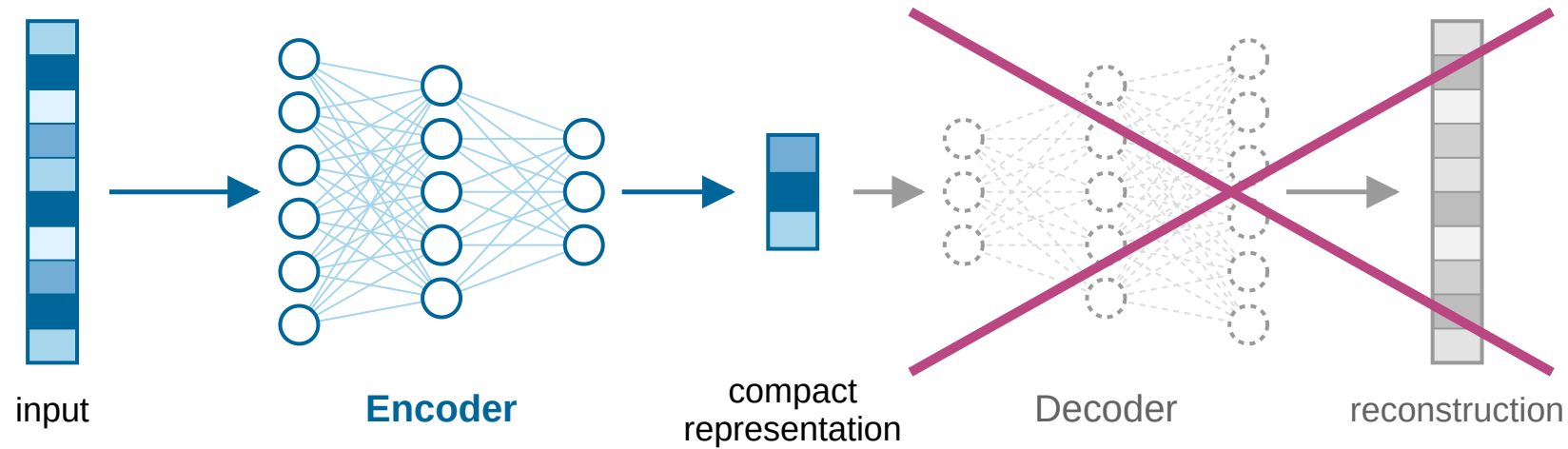
Encoder

Reduces the structure to a compact representation

Classifier

Assigns a symmetry label from that representation

Why not an autoencoder / VAE?



No decoder needed

The reconstruction capability is superfluous for our goal

Separate, not reconstruct

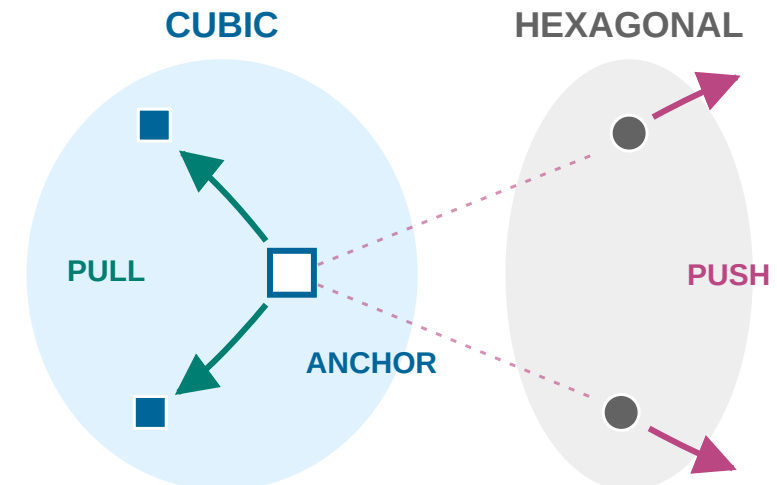
What matters is pulling different symmetries apart

How the SupCon loss works

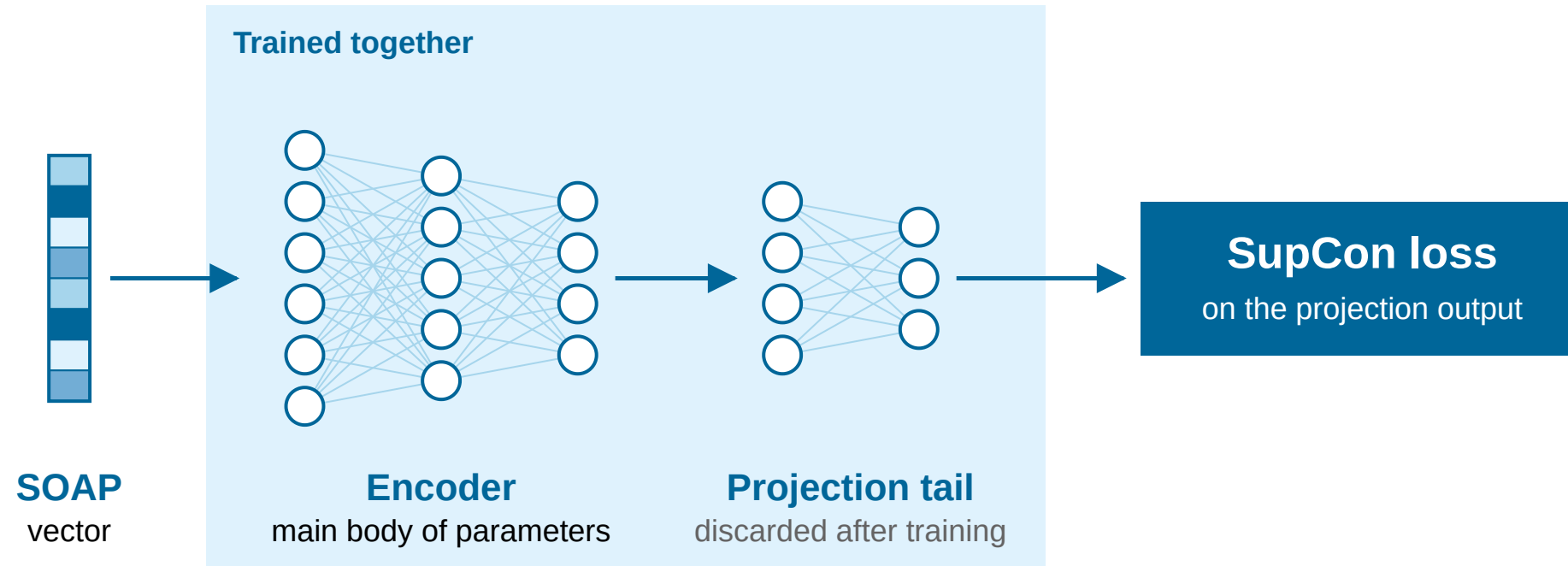
$$\mathcal{L} = \frac{1}{|B|} \sum_{i \in B} \left[-\frac{1}{|P(i)|} \sum_{p \in P(i)} \log \frac{e^{\text{sim}(\mathbf{z}_i, \mathbf{z}_p)/\tau}}{\sum_{a \in A(i)} e^{\text{sim}(\mathbf{z}_i, \mathbf{z}_a)/\tau}} \right]$$

$\text{sim}(\mathbf{z}_i, \mathbf{z}_j) = \frac{\mathbf{z}_i}{\|\mathbf{z}_i\|} \cdot \frac{\mathbf{z}_j}{\|\mathbf{z}_j\|}$ cosine similarity

$i \in B$	anchor: one structure of the batch
$P(i)$	positives: same symmetry as the anchor
$ P(i) $	number of positives of the anchor
$A(i)$	all other structures in the batch
τ	temperature: how sharp the softmax is

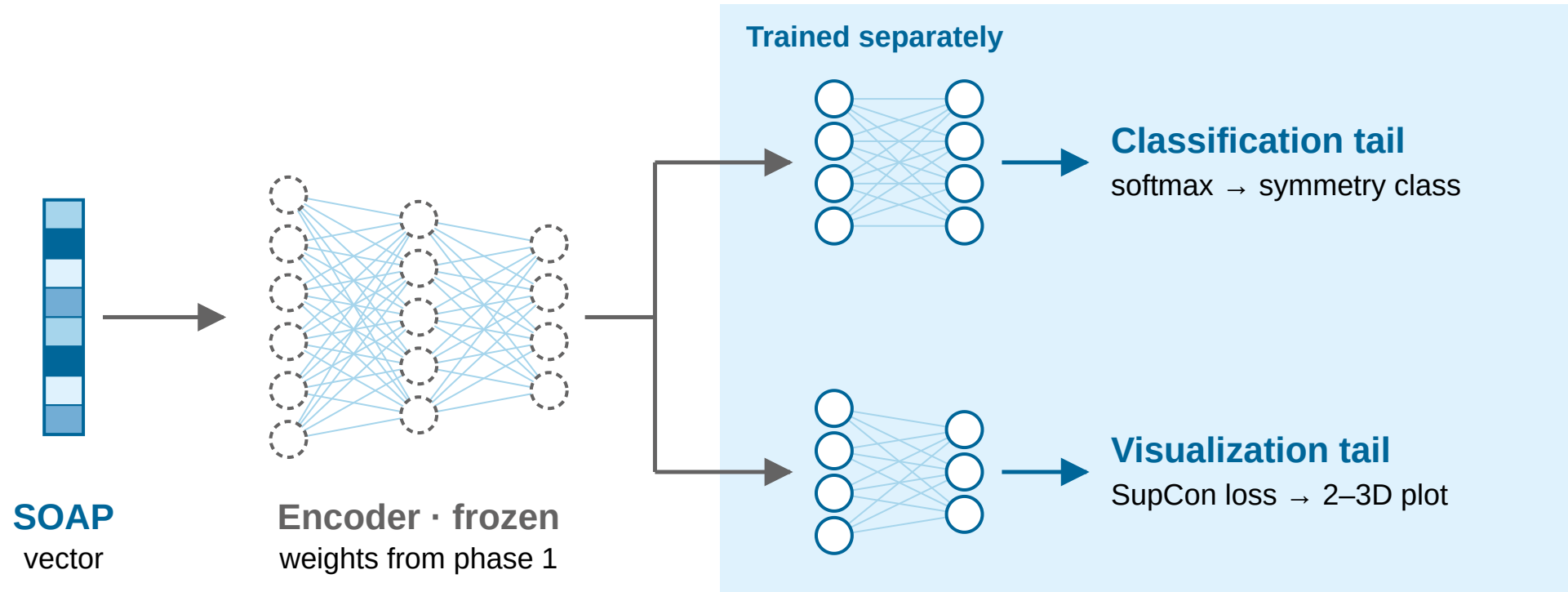


Architecture · Phase 1: learning the representation



All blocks are plain MLPs (Dense + ReLU) with different sizes

Architecture · Phase 2: frozen body, task tails



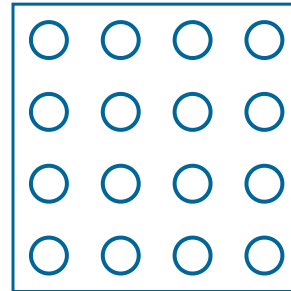
All blocks are plain MLPs (Dense + ReLU) with different sizes

Building the dataset: synthetic structures, known labels

Cubic
Hexagonal
Trigonal
Tetragonal
Orthorhombic
Monoclinic
Triclinic

Sample

a space group within each family



Generate

clean crystal with pyxtal

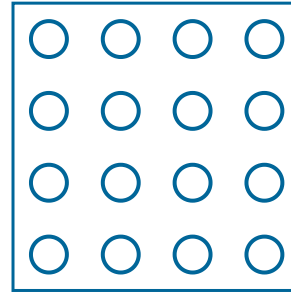


Building the dataset: synthetic structures, known labels

Cubic
Hexagonal
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Triclinic

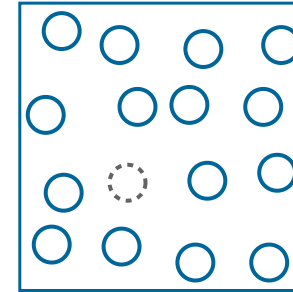
Sample

a space group within each family



Generate

clean crystal with pyxtal

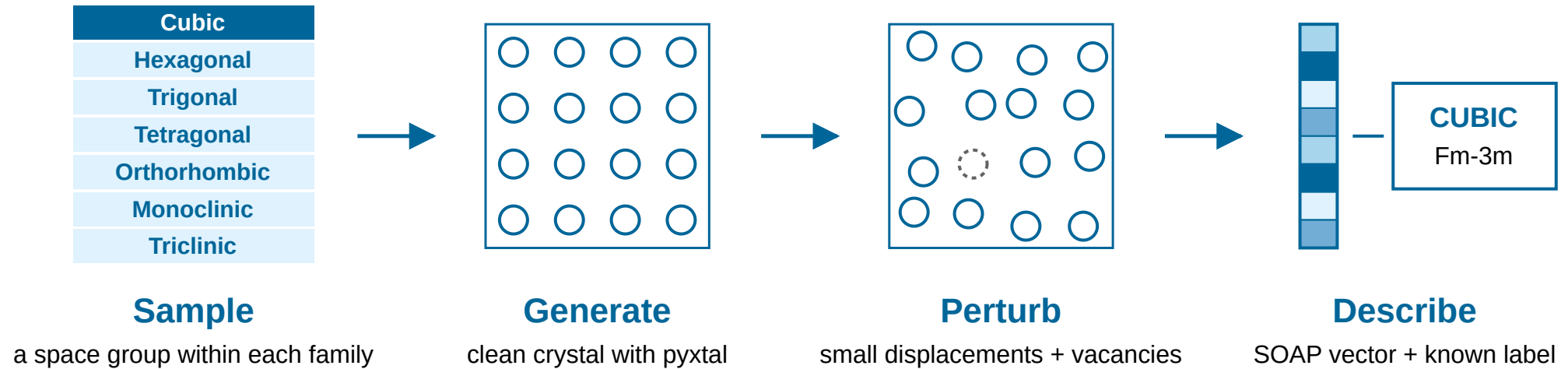


Perturb

small displacements + vacancies



Building the dataset: synthetic structures, known labels



Labels for free

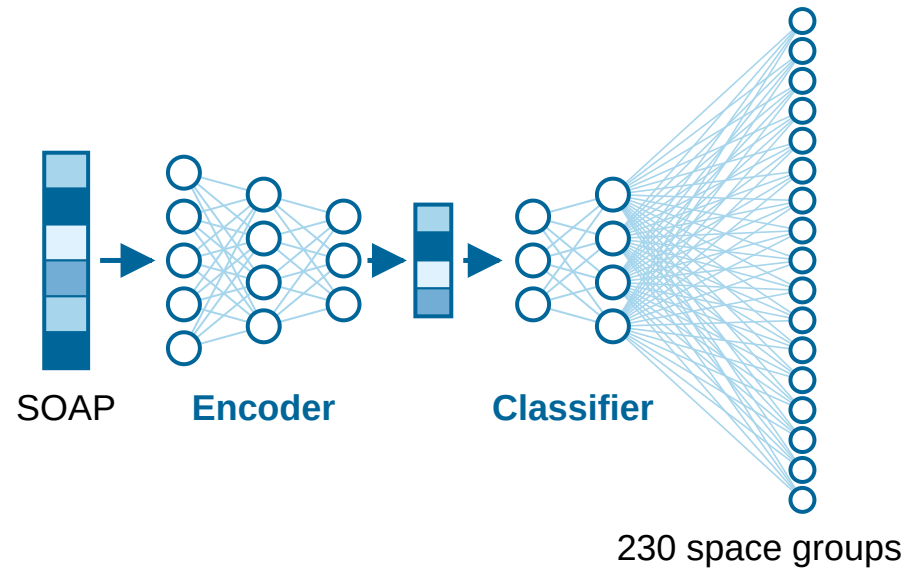
Family and space group are known by construction, even after the noise

Thermal noise

Perturbations mimic the disorder present in structures coming from high temperature simulations

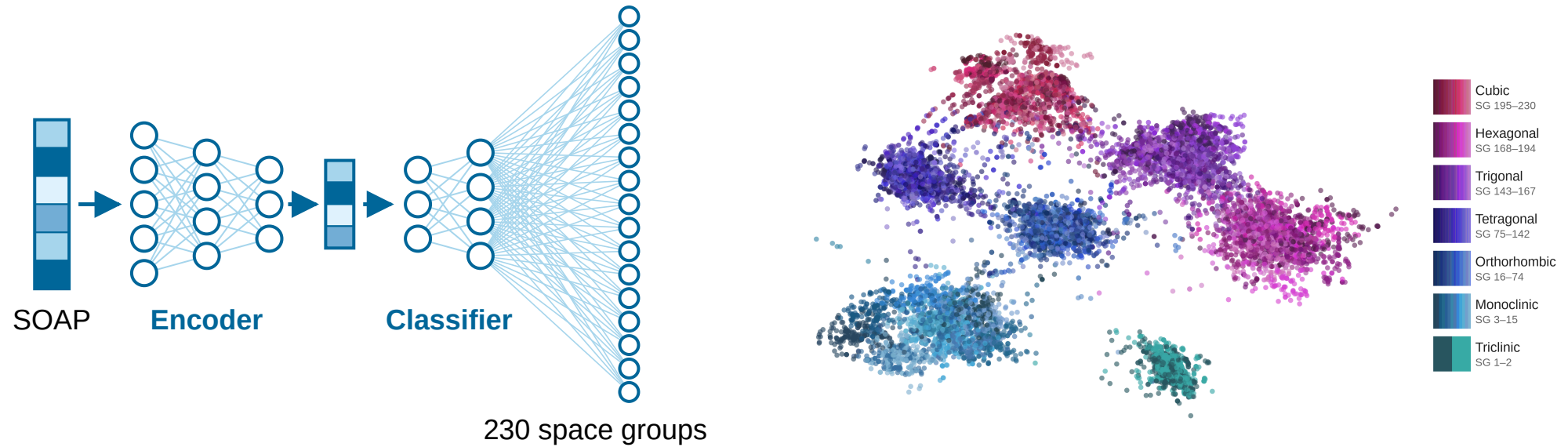
First obstacle: too many space groups at once

Single step



First obstacle: too many space groups at once

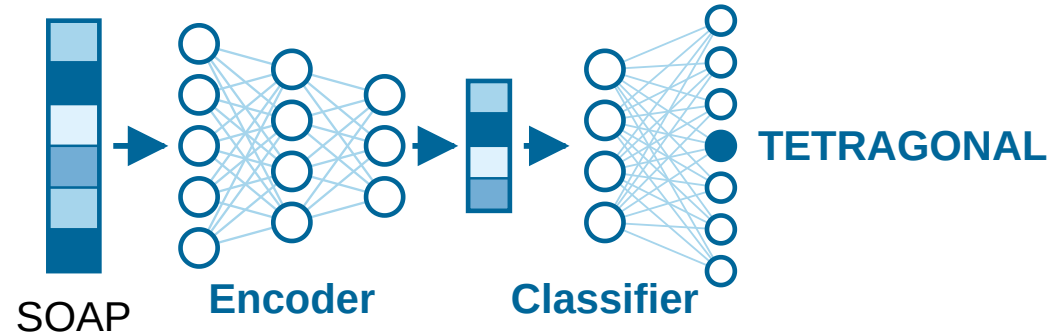
Single step



Did not work

One model for all 230 space groups has to face too much variety

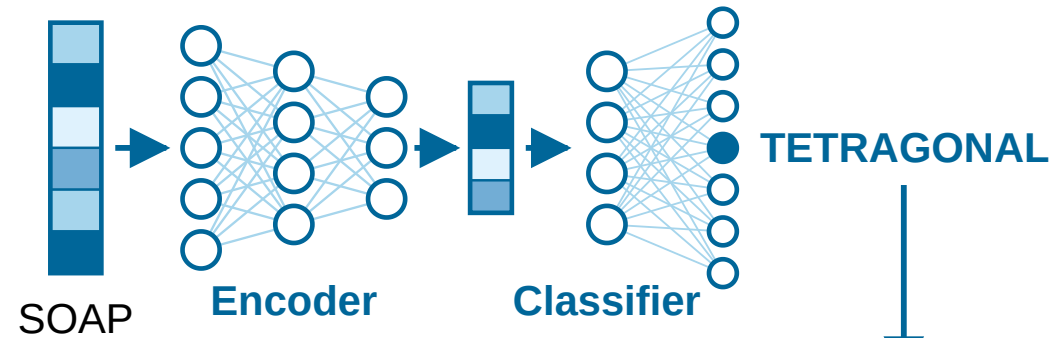
The fix: family first, then a space-group expert



Step 1 · Family

Predicts which of the 7 crystal families the structure belongs to

The fix: family first, then a space-group expert

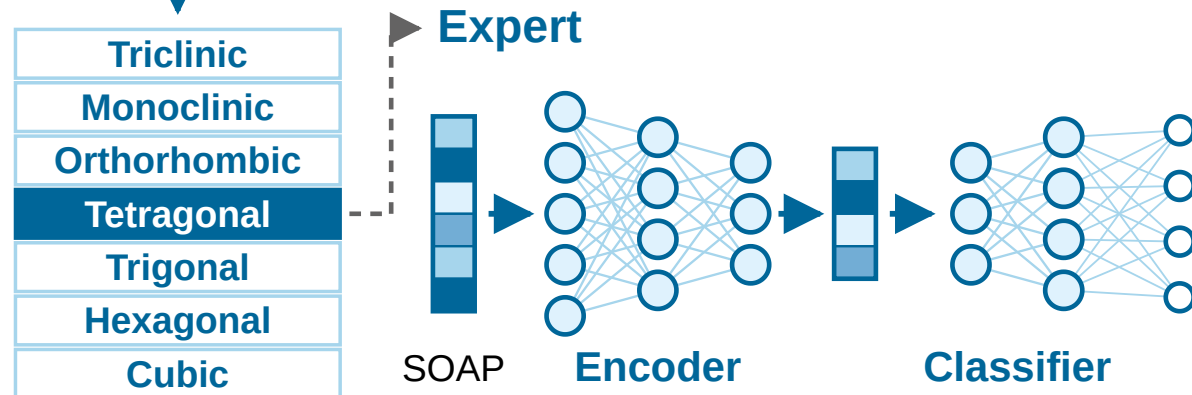


Step 1 - Family

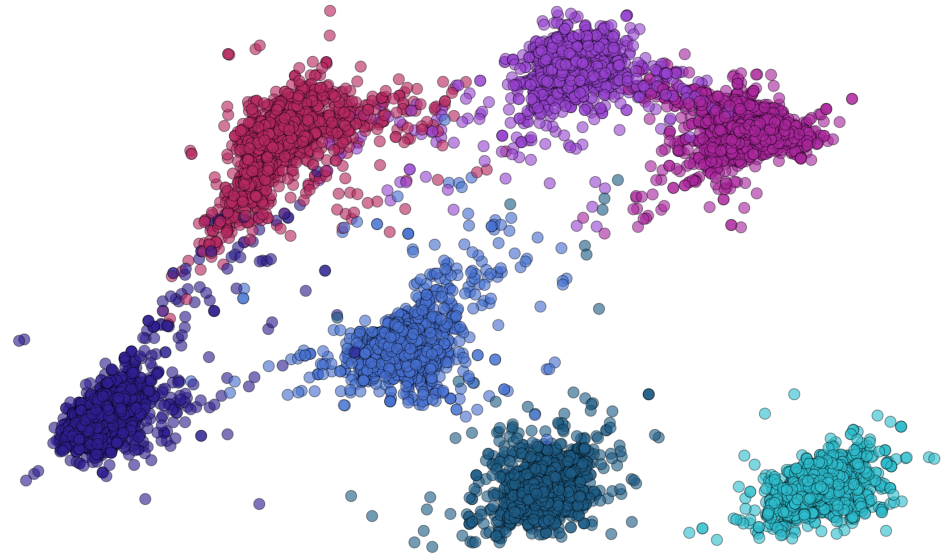
Predicts which of the 7 crystal families the structure belongs to

Step 2 - Space Group

The family expert has a smaller variance to face and now we predict the spacegroup



The family classifier: clear separation



2D map

Visualization tail → good separation by family

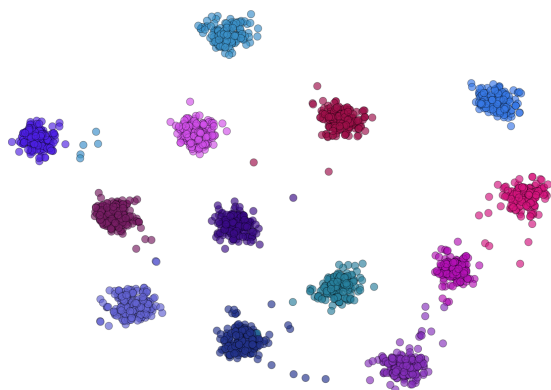
True family	Triclinic	99	<1		<1			
	Monoclinic	1	96	2	<1	<1	<1	<1
	Orthorhombic	<1	4	92	1	<1	1	1
	Tetragonal			1	96	<1		2
	Trigonal			1		90	8	2
	Hexagonal			<1		5	95	
	Cubic				1	1		98
			Triclinic	Monoclinic	Orthorhombic	Tetragonal	Trigonal	Hexagonal
		Predicted family						

Confusion matrix

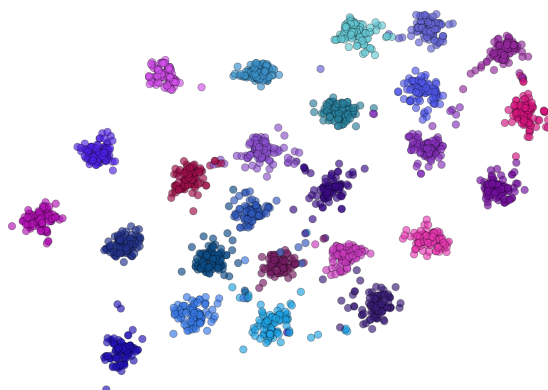
Strongly diagonal, mean accuracy ≈ 95%

One space-group expert per family

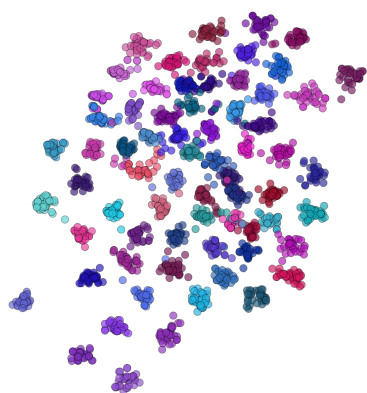
Monoclinic (13 space groups)



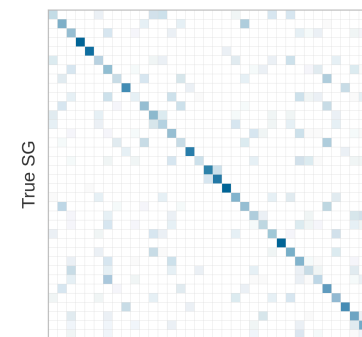
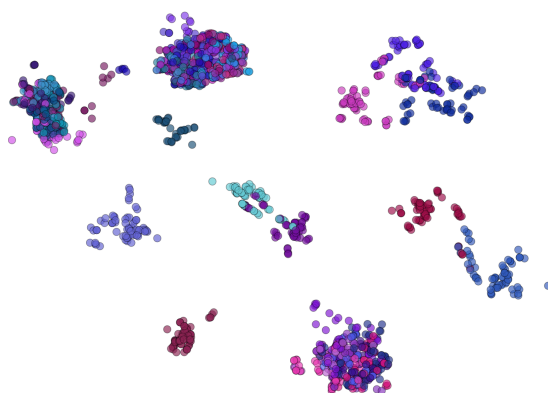
Trigonal (25 space groups)



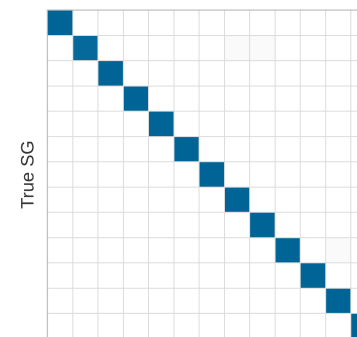
Tetragonal (68 space groups)



Cubic (36 space groups)



Cubic



Monoclinic

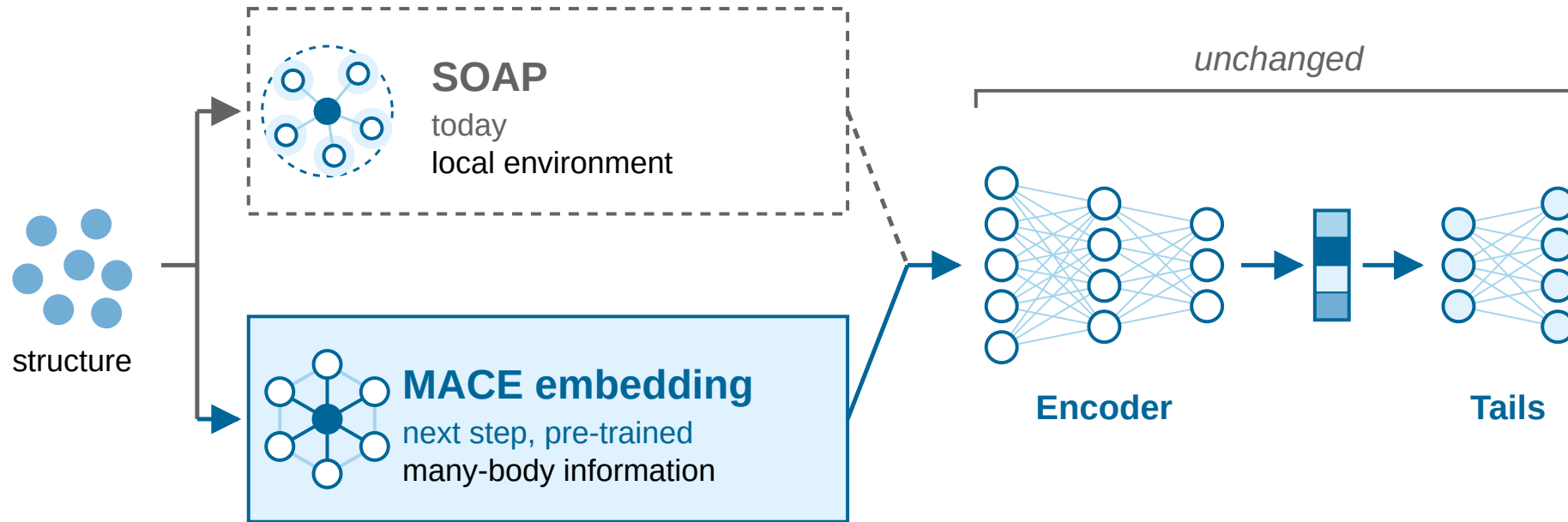
Separation

Clear for all families except Cubic, as the confusion matrices confirm

Suspected reason

Probably the input (SOAP), not the model; still under investigation

Next step: MACE embedding instead of SOAP



Only the input changes: a hypothesis to test, not yet run

Conclusions

Problem

Nested Sampling structures are too noisy for classical symmetry tools



Approach

SupCon on SOAP: family first, then a space-group expert



Status

Clear separation by family; Cubic experts still open



Next step

Replace SOAP + encoder with a pre-trained MACE embedding

Thank you

Questions and feedback are welcome